

ECE 253 – Digital Image Processing

Image Restoration

Model of Image Degradation & Restoration

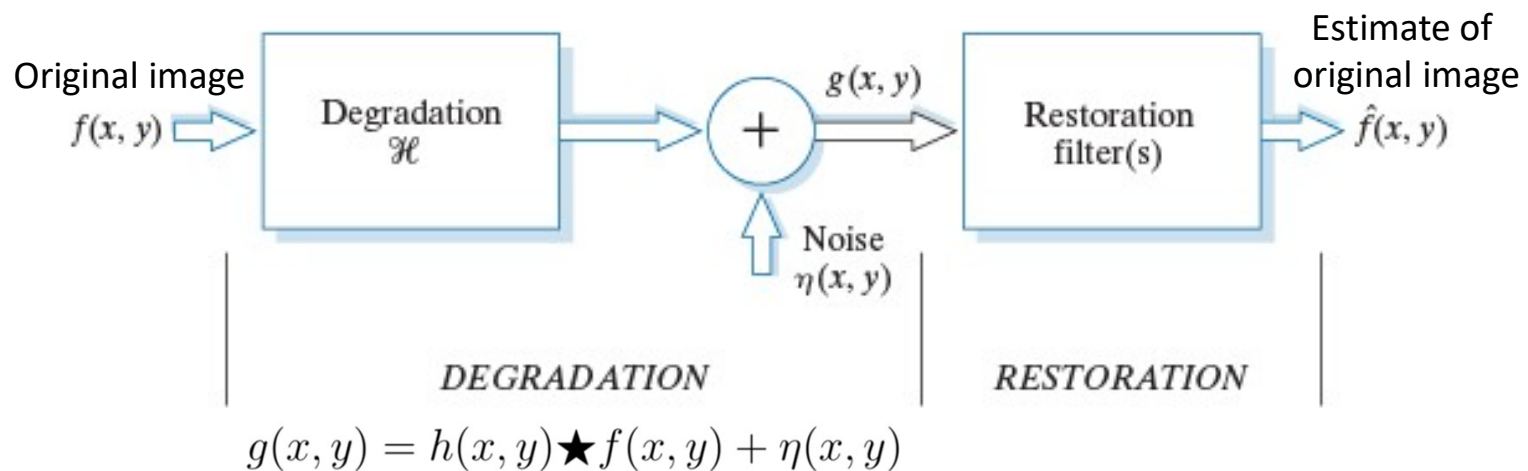
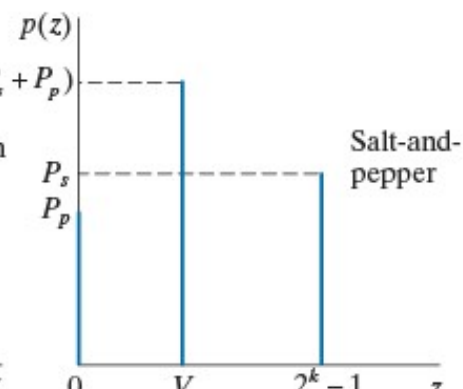
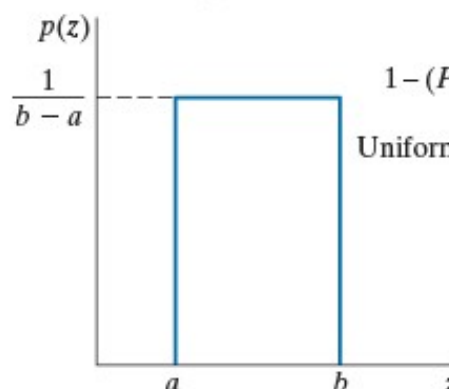
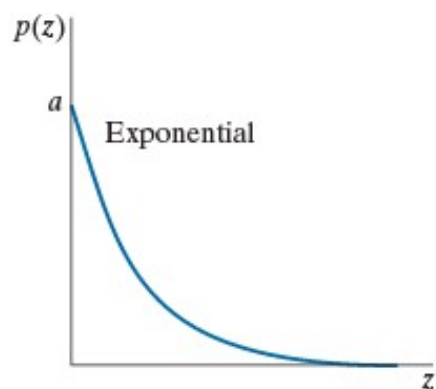
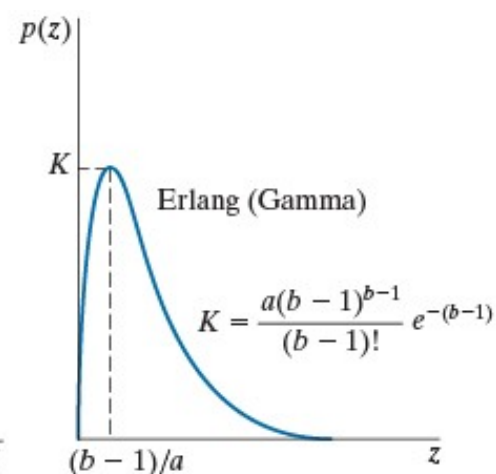
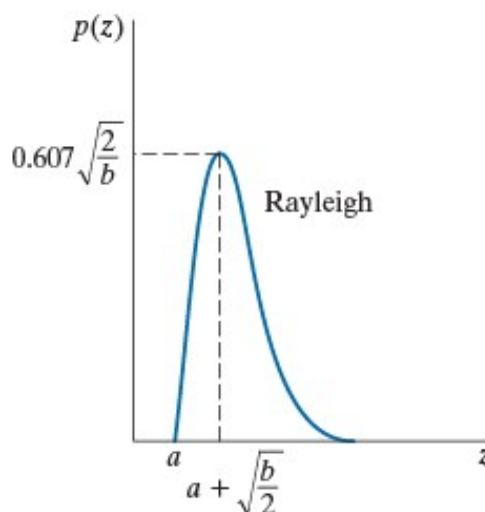
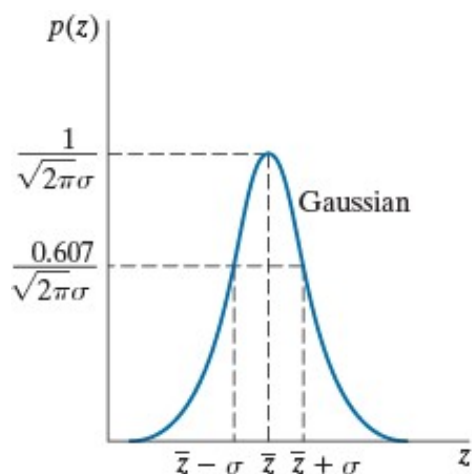


Image Restoration

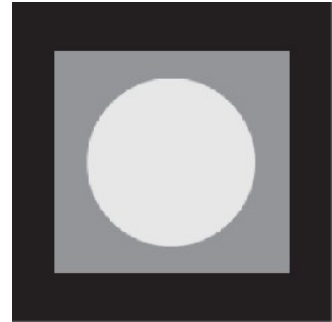
Remove noise. Estimate original image using Deconvolution

Noise: Probability Density Functions



Adding noise from different models

Original Image
 $f(x, y)$



$g(x, y)$

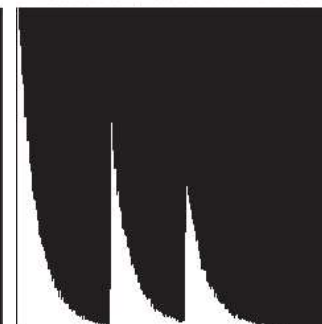
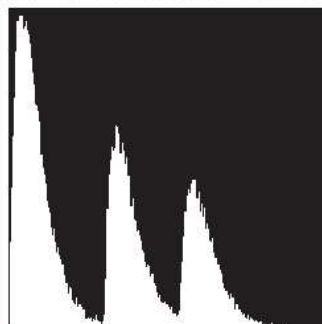
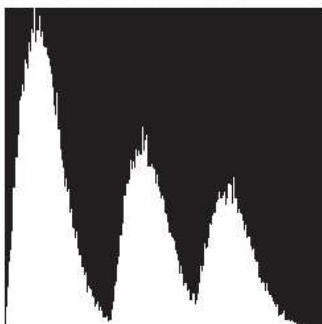
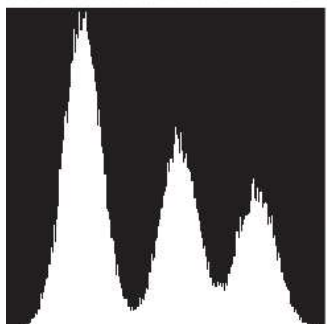
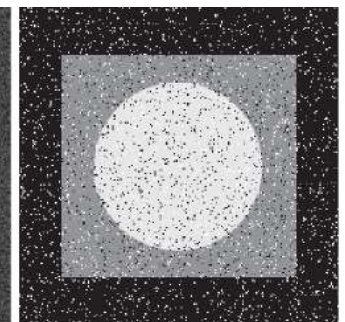
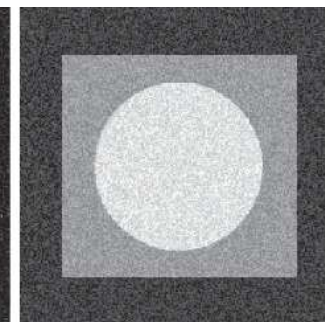
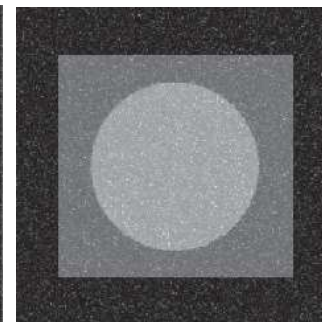
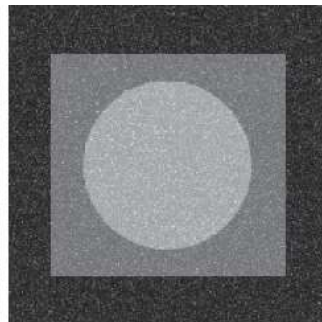
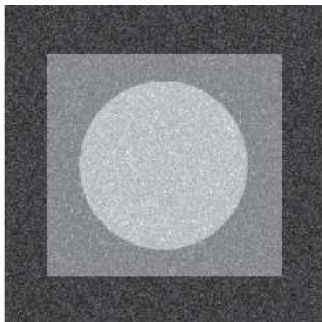
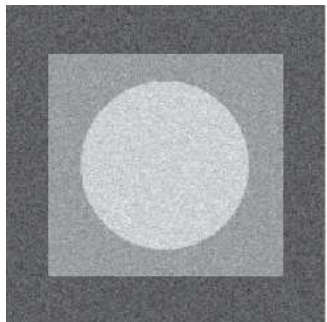
$g(x, y)$

$g(x, y)$

$g(x, y)$

$g(x, y)$

$g(x, y)$



Gaussian

Rayleigh

Gamma

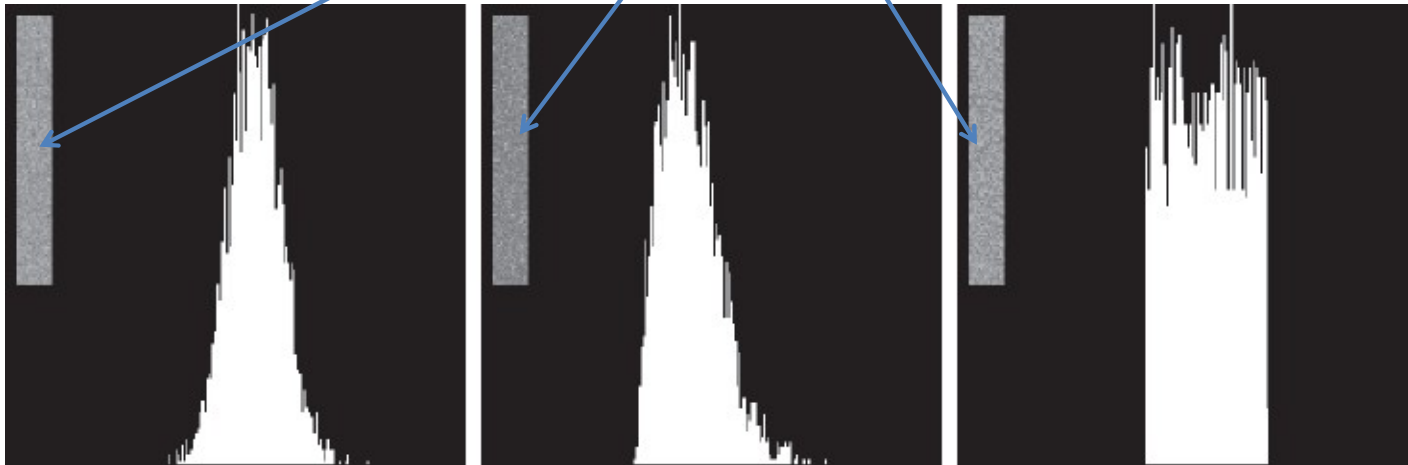
Exponential

Uniform

Salt & pepper

Histograms of sample patches

1. Sample “flat” patches from images with noise



2. Identify closest probability density function (pdf) match

Gaussian

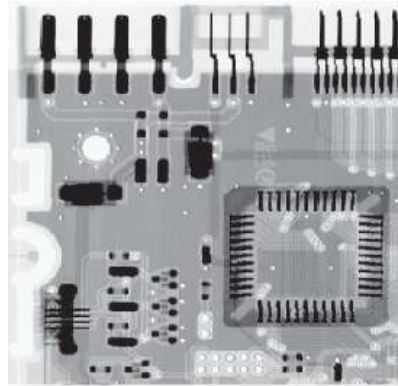
Rayleigh

Uniform

Mean filters

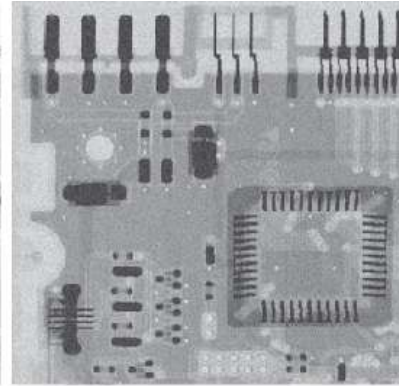
X-ray
image

$$f(x, y)$$

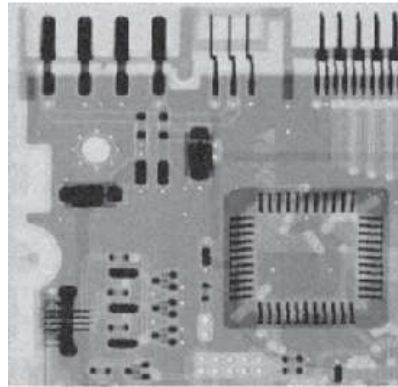


Additive
Gaussian
noise

$$g(x, y)$$

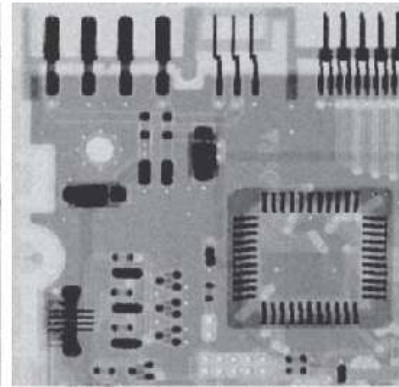


Arithmetic
mean
filtered



$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r,c) \in S_{xy}} g(r, c)$$

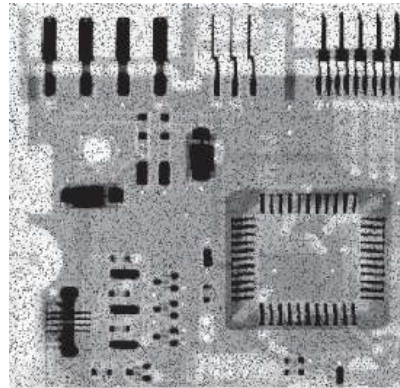
Geometric
mean
filtered



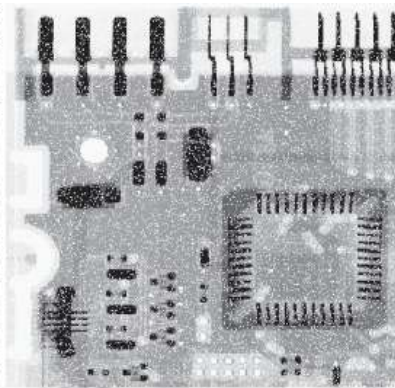
$$\hat{f}(x, y) = \left[\prod_{(r,c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$

Mean filters

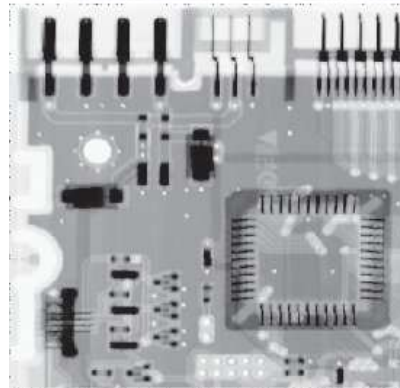
Additive
pepper
noise
 $g(x, y)$



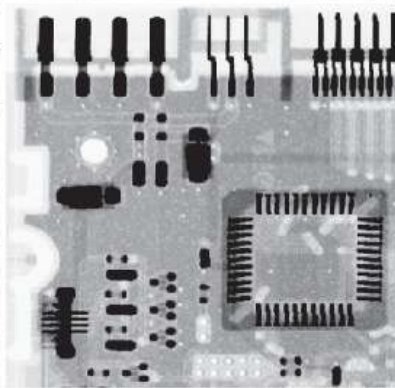
Additive
salt
noise
 $g(x, y)$



Contraharmonic
mean filtered
 $\hat{f}(x, y)$



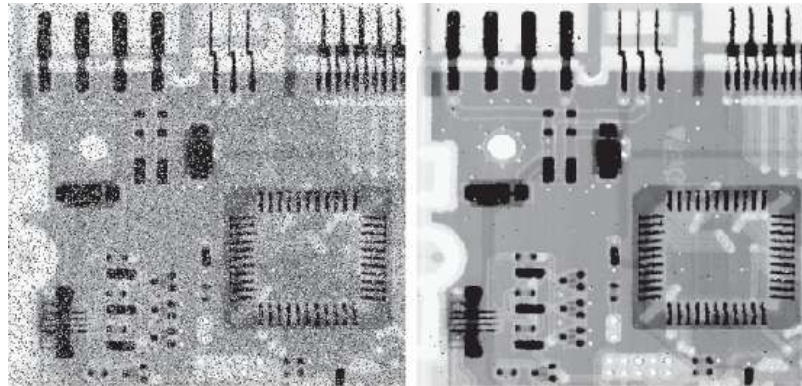
Contraharmonic
mean filtered
 $\hat{f}(x, y)$



Order-statistic filters

Additive
salt and
pepper
noise

$g(x, y)$

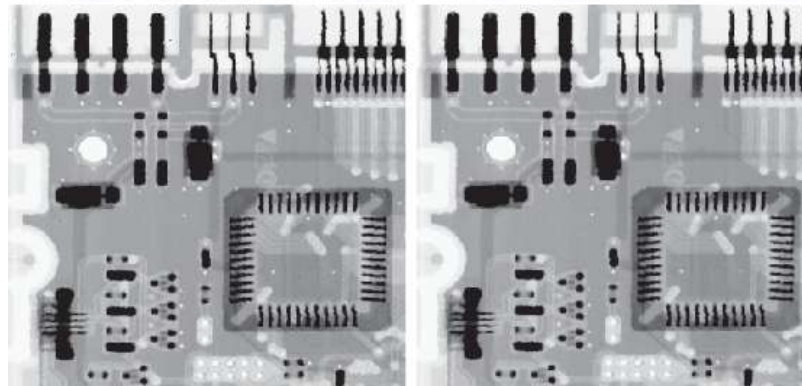


1x median
filtered

$$\hat{f}(x, y) = \operatorname{median}_{(r, c) \in S_{xy}} \{g(r, c)\}$$

2x median
filtered

$\hat{f}(x, y)$



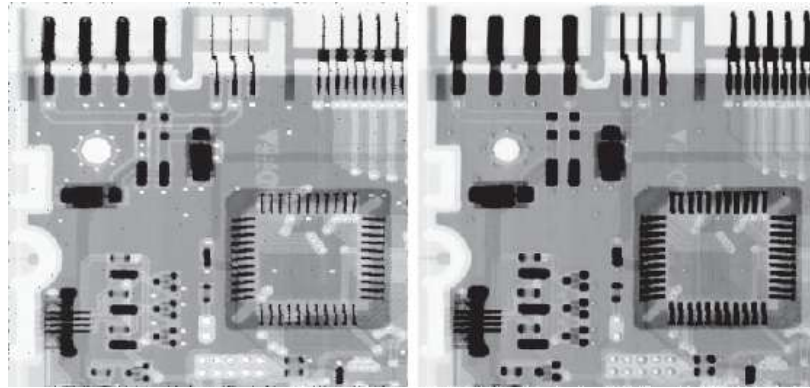
3x median
filtered

$\hat{f}(x, y)$

Order-statistic filters

Max
filtered

$$\hat{f}(x, y)$$

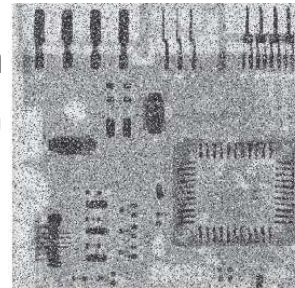


Min
filtered

$$\hat{f}(x, y)$$

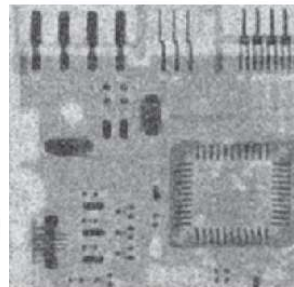
Comparing filters

Additive uniform
+ $g(x, y)$
salt and pepper
noise



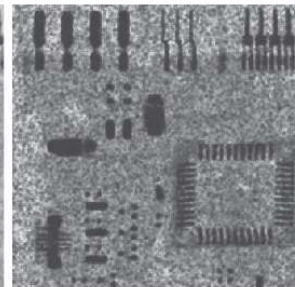
Arithmetic mean
filtered

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} g(r, c)$$



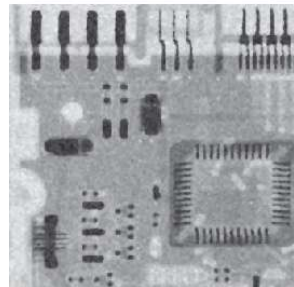
Geometric mean

$$\hat{f}(x, y) = \left[\prod_{(r, c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$



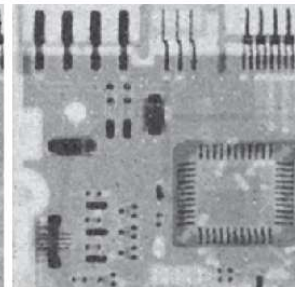
Median
filtered

$$\hat{f}(x, y) = \text{median} \{g(r, c)\}_{(r, c) \in S_{xy}}$$



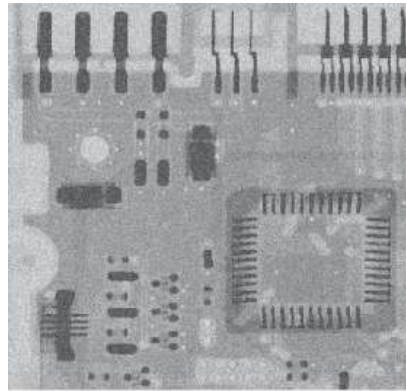
Alpha-trimmed
mean filtered

$$\hat{f}(x, y)$$



Adaptive filters

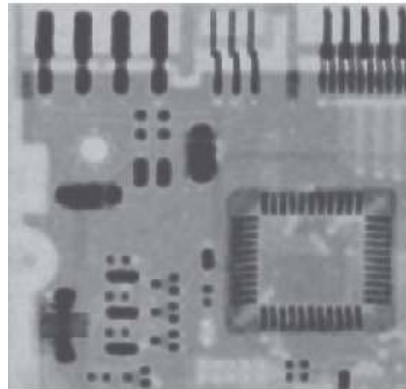
Additive
Gaussian
noise
 $g(x, y)$



Arithmetic
mean filtered

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r,c) \in S_{xy}} g(r, c)$$

Geometric
mean
filtered



Adaptive noise
reduction
filtered

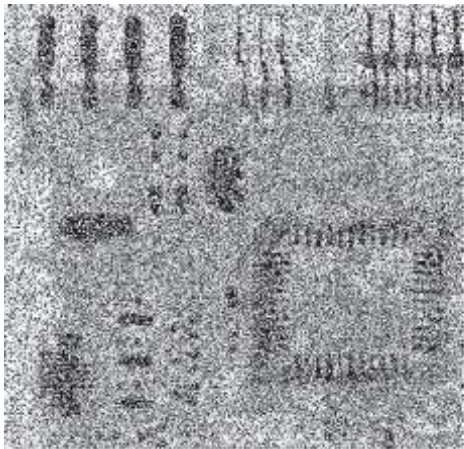
$$\hat{f}(x, y) = \left[\prod_{(r,c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$

$$\hat{f}(x, y) = g(x, y) - \min \left(1, \frac{\sigma_{\eta}^2}{\sigma_{S_{xy}}^2} \right) [g(x, y) - \bar{z}_{S_{xy}}]$$

Input: Overall noise variance σ_{η}^2
 Local variance $\sigma_{S_{xy}}^2$ (computed)
 Local mean $\bar{z}_{S_{xy}}$

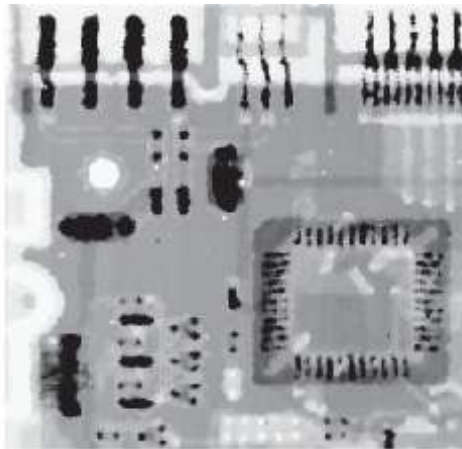
Adaptive filters

Additive
salt and pepper
noise



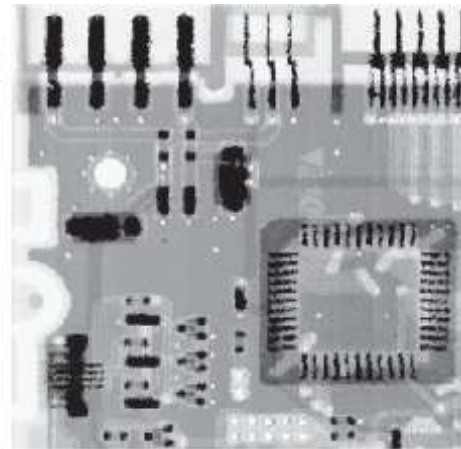
$g(x, y)$

Median
filtered



$\hat{f}(x, y) = \text{median}_{(r,c) \in S_{xy}} \{g(r, c)\}$

Adaptive median
filtered



$\hat{f}(x, y)$

Changes size of
neighborhood according to
noise density

Model of image degradation

Spatial domain

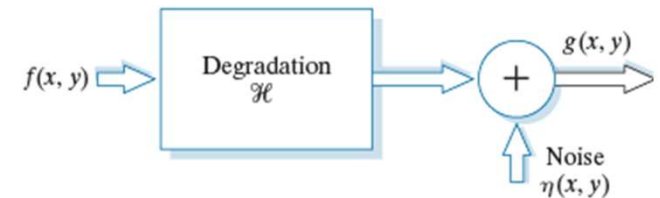
$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y)$$

Degraded
image

Degradation
function

Original
image

Noise
image



Frequency domain

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Degraded
image

Degradation
function

Original
image

Noise
image

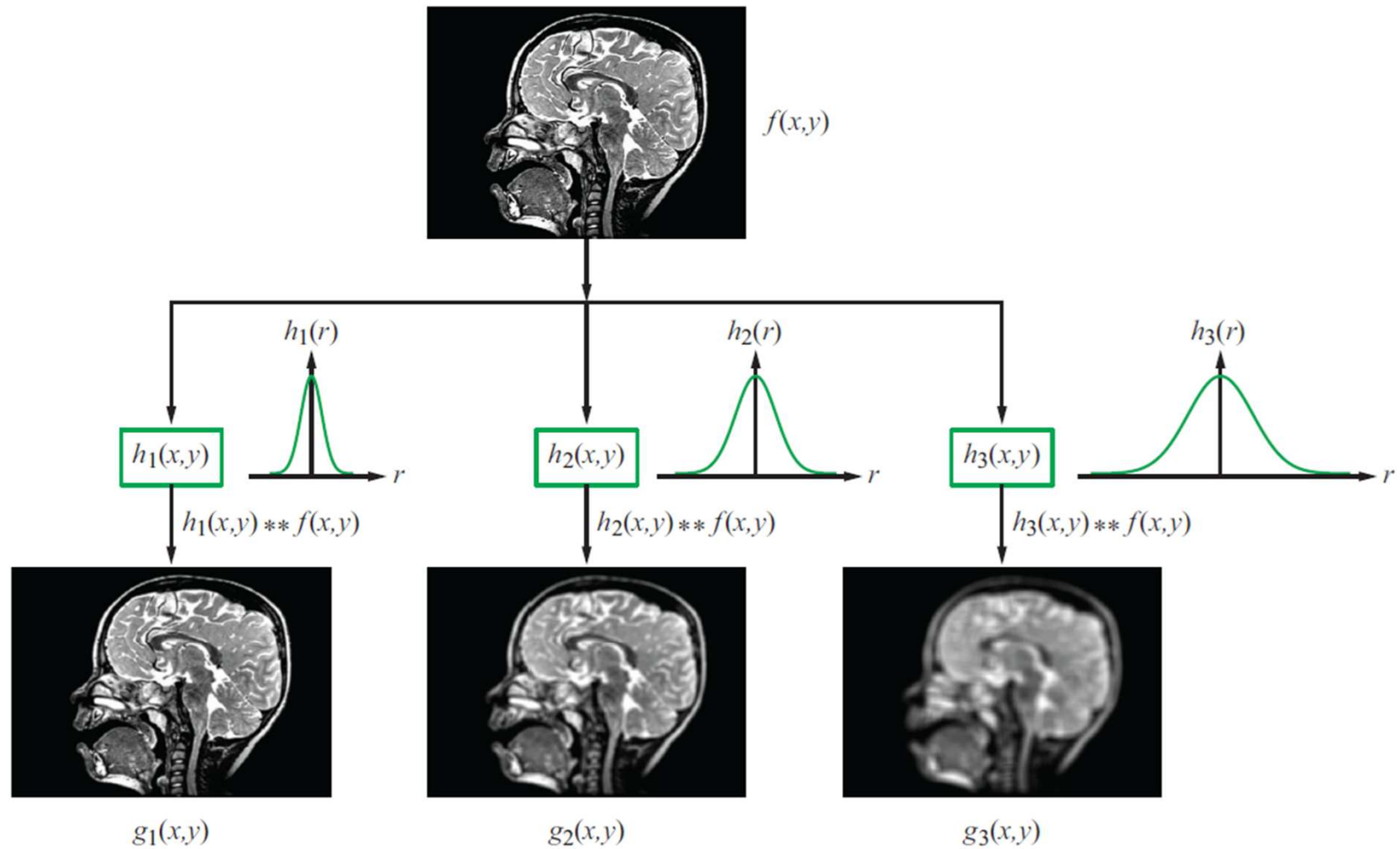


Figure 6-1 Simulation of image blurring: the original noise-free image at the top is convolved with Gaussian-shaped point spread functions of different effective widths. The variable r is the radial distance $r = \sqrt{x^2 + y^2}$.

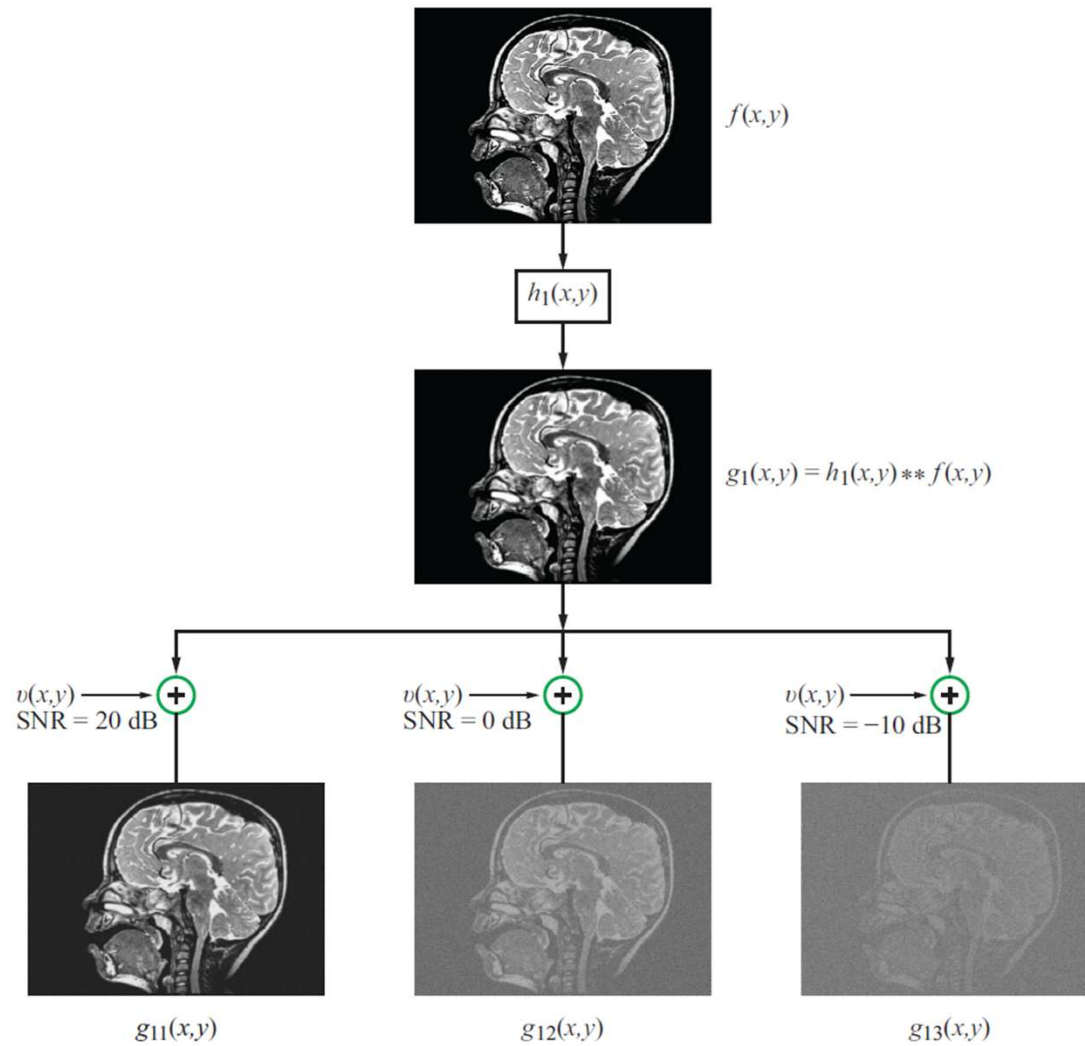


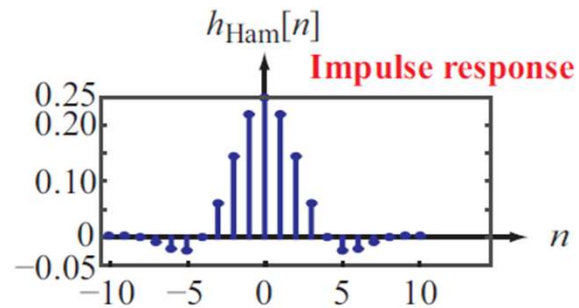
Figure 6-2 The image in the upper center, $g_1(x,y)$, had been convolved with the narrow filter before noise was added to it. $\text{SNR} = 20 \text{ dB}$ corresponds to average signal power/average noise power = 100, so $g_{11}(x,y)$ is essentially noise-free. In contrast, $\text{SNR} = 0 \text{ dB}$ in $g_{12}(x,y)$, which means that the average signal and noise powers are equal, and in $g_{13}(x,y)$ the noise power is $10\times$ the signal power.

$$\mathbf{H}_{\text{brick}}(\Omega_1, \Omega_2) = \begin{cases} 1 & \text{for } 0 \leq \Omega_1, \Omega_2 \leq \Omega_c, \\ 0 & \text{for } \Omega_c < \Omega_1, \Omega_2 \leq \pi. \end{cases}$$

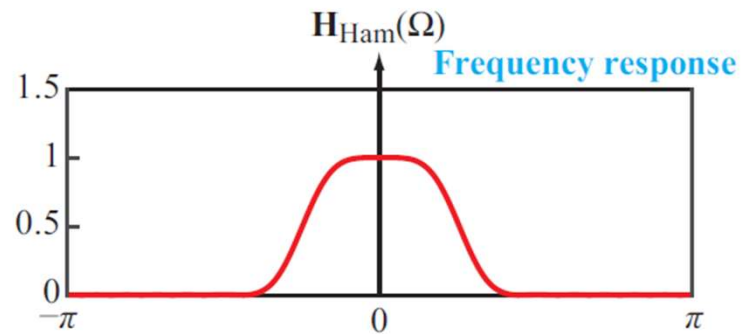
Lowpass filtering: formulae for brickwall and Hamming filters.

$$\begin{aligned} h_{\text{brick}}[n, m] &= h_{\text{brick}}[n] h_{\text{brick}}[m] \\ &= \left(\frac{\Omega_c}{\pi} \right)^2 \text{sinc} \left(\frac{\Omega_c n}{\pi} \right) \text{sinc} \left(\frac{\Omega_c m}{\pi} \right) \end{aligned}$$

$$\begin{aligned} h_{\text{Ham}}[n, m] &= h_{\text{Ham}}[n] h_{\text{Ham}}[m] = \\ &\begin{cases} \left(\frac{\Omega_c}{\pi} \right)^2 \text{sinc} \left(\frac{\Omega_c n}{\pi} \right) \left[0.54 + 0.46 \cos \left(\frac{\pi n}{N} \right) \right] \\ \quad \times \text{sinc} \left(\frac{\Omega_c m}{\pi} \right) \left[0.54 + 0.46 \cos \left(\frac{\pi m}{N} \right) \right] & \text{for } |n|, |m| \leq N, \\ 0 & \text{for } |n|, |m| > N. \end{cases} \end{aligned}$$

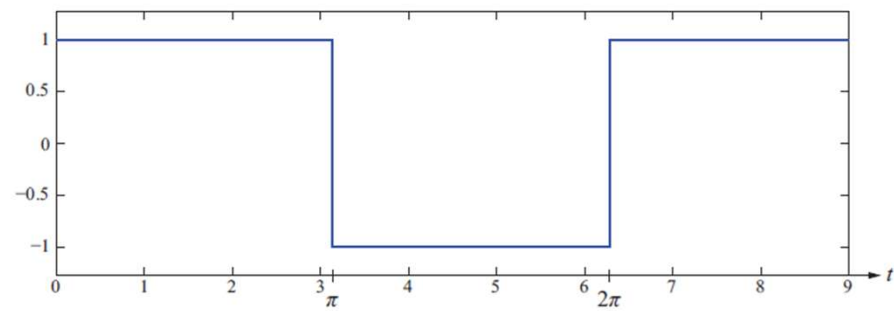


(a) Impulse response of Hamming-windowed filter

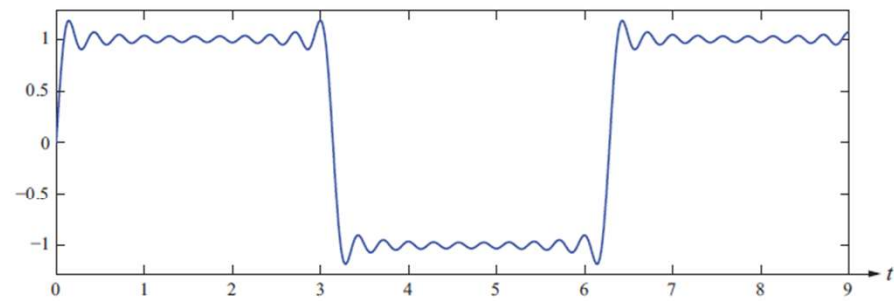


(b) Spectrum $\mathbf{H}_{\text{Ham}}(\Omega)$ of Hamming-windowed filter

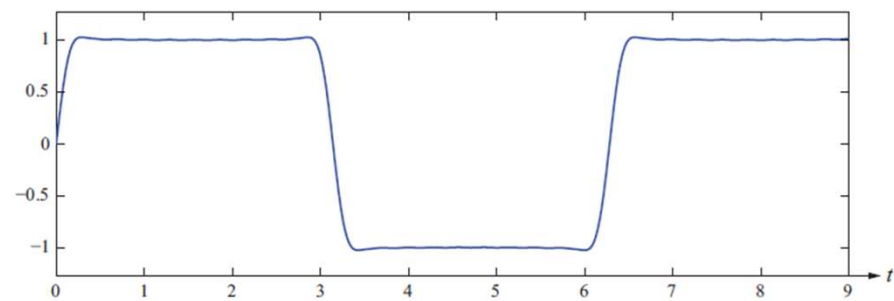
Figure 6-5 Hamming window of length $N = 10$: (a) impulse response, and (b) spectrum.



(a) Square wave $x(t)$



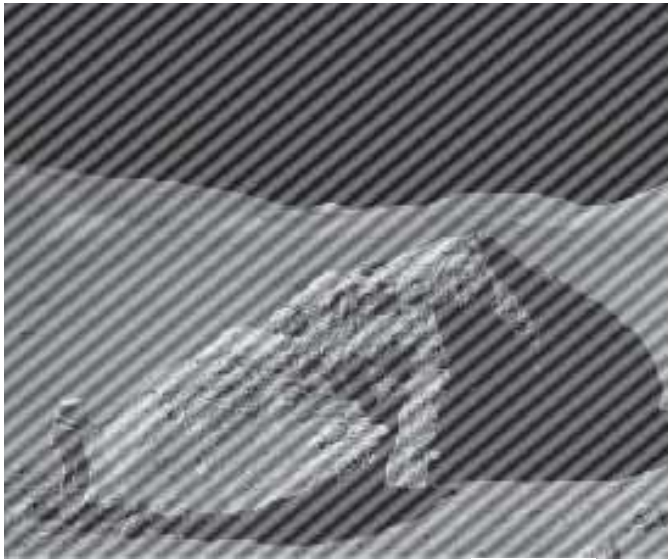
(b) Brickwall lowpass-filtered signal $y_{\text{brick}}(t)$



(c) Tapered Fourier series signal $y_{\text{Ham}}(t)$

Figure 6-4 (a) Square wave $x(t)$, (b) brickwall lowpass-filtered version, and (c) Hamming-windowed version.

Periodic noise

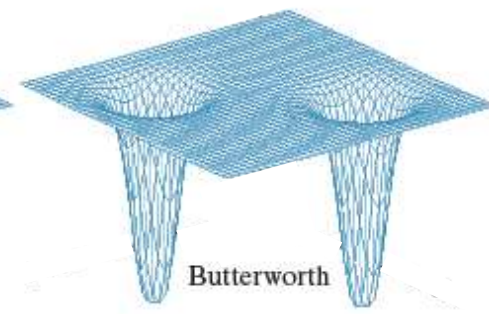
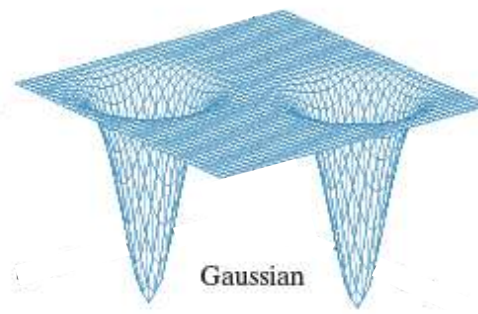
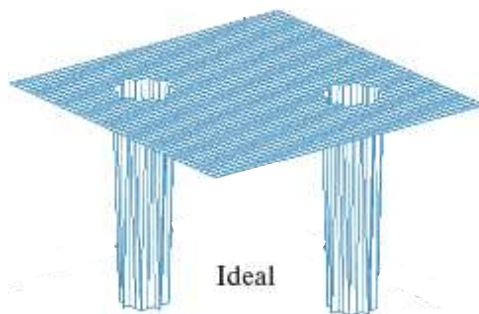


Additive sinusoidal noise $g(x, y)$



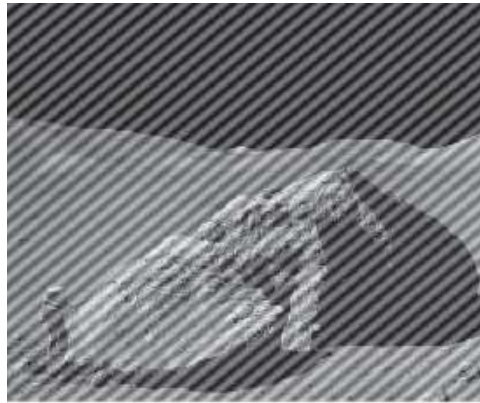
DFT Magnitude. Conjugate impulses

Notch Filter: Band
of frequencies,
specific
orientation(s)

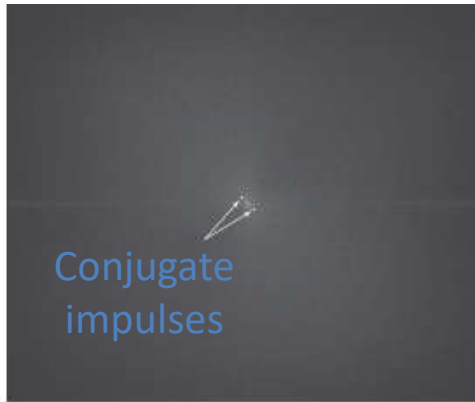


Notch Filter

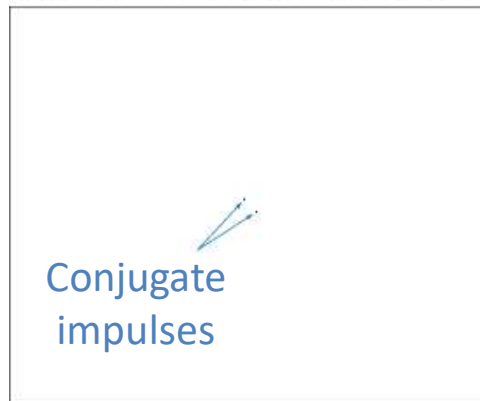
Degraded image
 $g(x, y)$



DFT
magnitude



Filter in
frequency
domain



Estimate
of original
image
 $\hat{f}(x, y)$



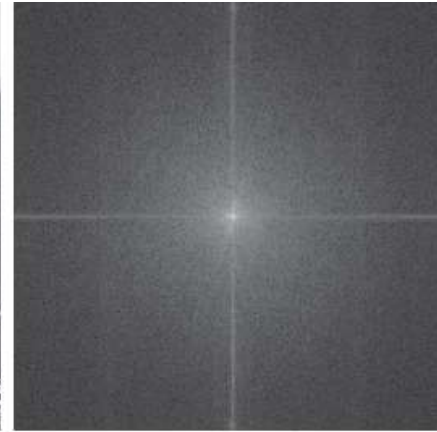
Notch reject filter

Degraded
image

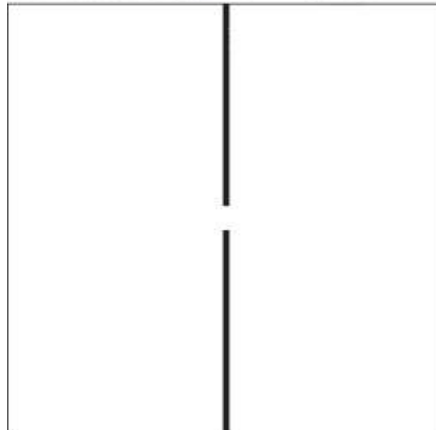
$$g(x, y)$$



DFT
magnitude

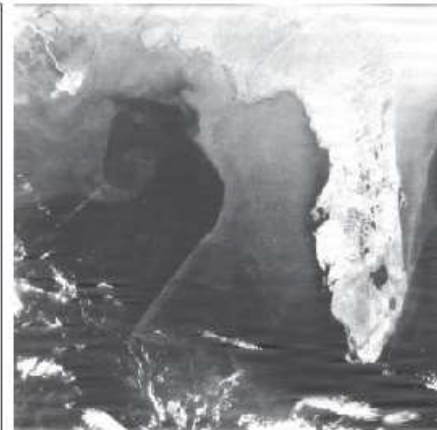


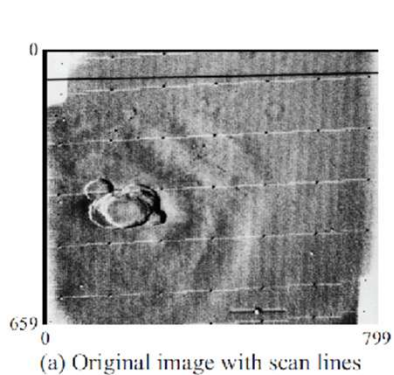
Filter in
frequency
domain



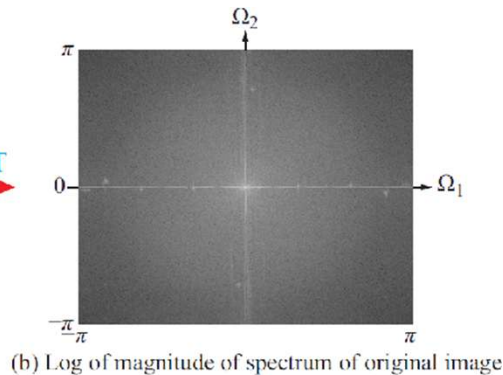
Estimate
of original
image

$$\hat{f}(x, y)$$



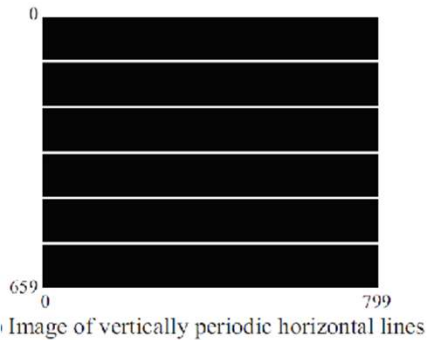


DSFT
→

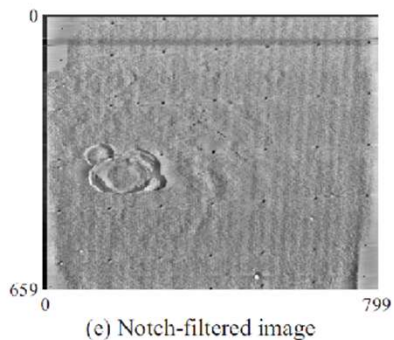
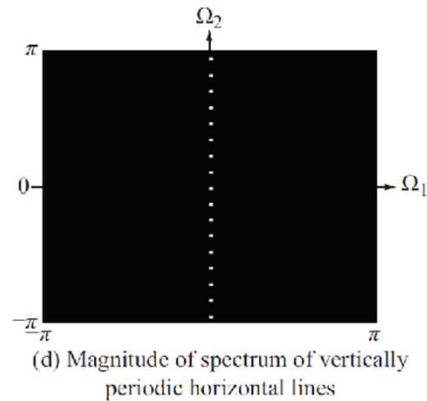


Notch Filtering:
Remove sinusoidal
Interference.

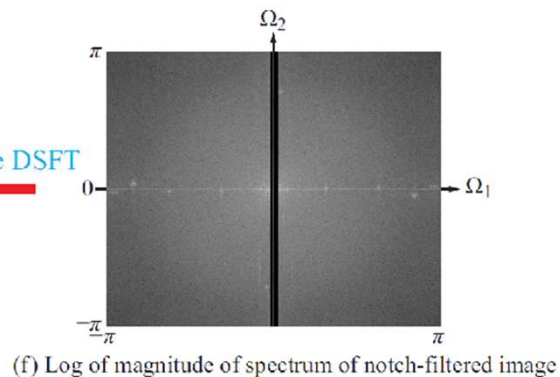
Here: using notch
filtering to remove
scan lines from
Mariner image.



DSFT
→



Inverse DSFT
←

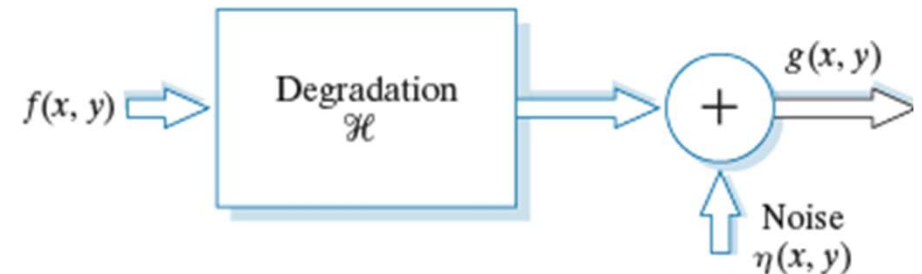


Horizontal scan lines
appear in vertical axis
of spectrum. Eliminate
these eliminates the
scan lines.

Estimating the degradation function

Methods

- Observation
- Experimentation
- Mathematical modeling



Observation

Manually process a subimage $g_s(x, y)$ of the degraded image $g(x, y)$ and let the result be our estimate of the original subimage $\hat{f}_s(x, y)$

$$g_s(x, y) = h_s(x, y) \star \hat{f}_s(x, y), \text{ solve for } h_s(x, y)$$

This is called blind deconvolution because the true degradation function is not known.
Solve in the frequency domain

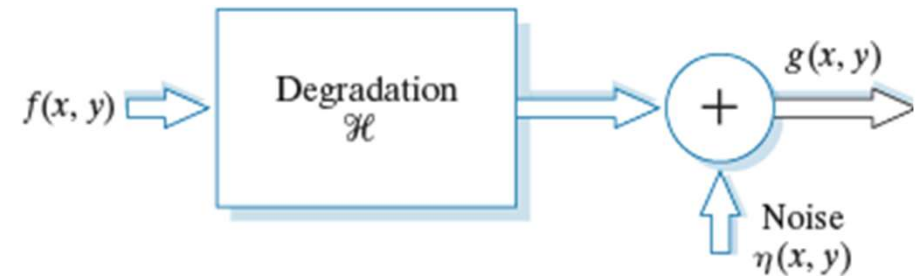
$$G_s(u, v) = H_s(u, v) \hat{F}_s(u, v)$$
$$H_s(u, v) = \frac{G_s(u, v)}{\hat{F}_s(u, v)}$$

then construct $H(u, v)$ for whole image with the same basic shape as $H_s(u, v)$

Estimating the degradation function

Methods

- Observation
- Experimentation
- Mathematical modeling



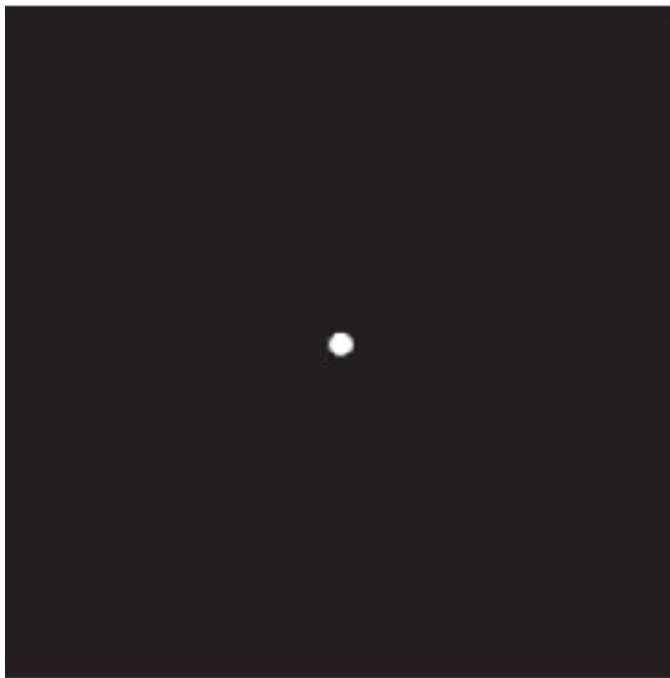
Experimentation

If the imaging system is available, then obtain the impulse response of the degradation by imaging an impulse (i.e., a small dot of light, as bright as possible)

$$H(u, v) = \frac{G(u, v)}{A}, \text{ where } A \text{ is a constant describing the strength of the impulse}$$

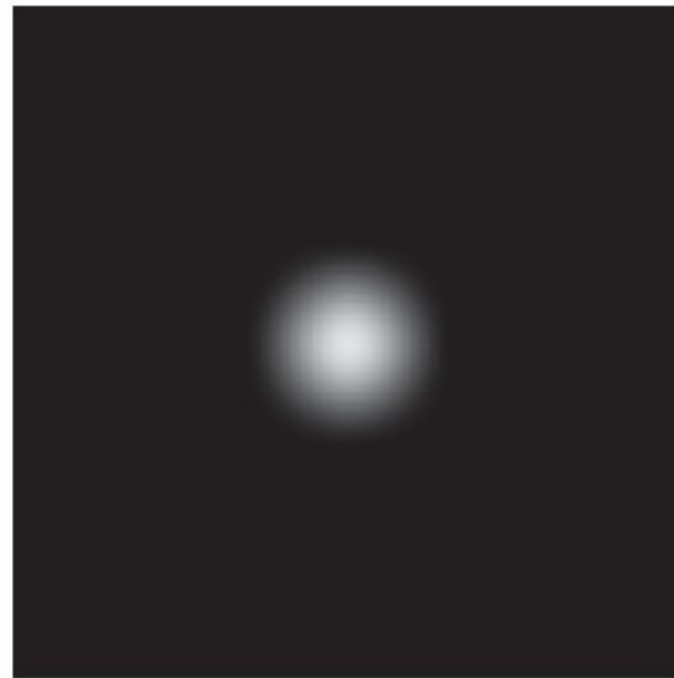
DFT of acquired, degraded image

Estimation of degradation function by experimentation



Impulse of light

$$f(x, y)$$



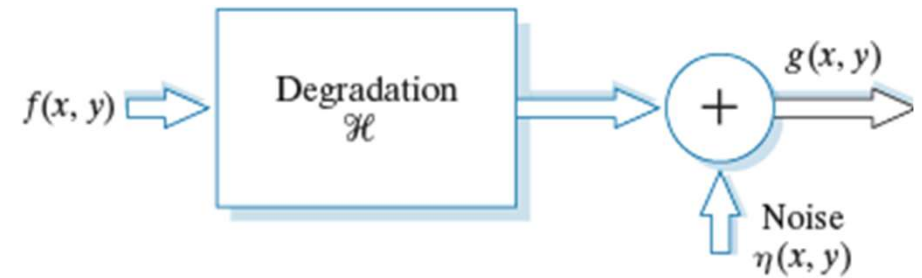
Imaged (degraded) impulse

$$g(x, y)$$

Estimating the degradation function

Methods

- Observation
- Experimentation
- Mathematical modeling



Mathematical Modeling

Model $H(u, v)$ known parameters

Estimation of degradation function by mathematical modeling

$f(x, y)$



Severe turbulence
 $g(x, y)$



Atmospheric
turbulence
model

Mild turbulence

$g(x, y)$



Low turbulence
 $g(x, y)$



Image Restoration

Inverse filtering

$$G(u, v) = H(u, v)\hat{F}(u, v)$$

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)}$$

– Mitigate divide by zero

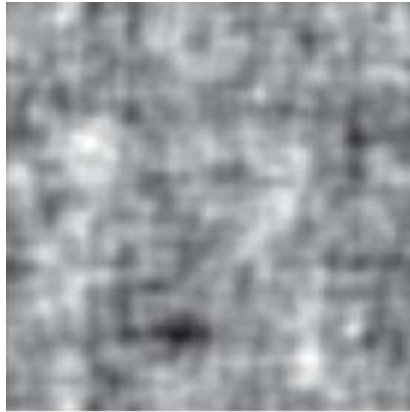
- Threshold $H(u, v)$
- Ideal lowpass filter $H(u, v)$

$$\hat{F}(u, v) = \begin{cases} \frac{G(u, v)}{H(u, v)} & \text{if } |H(u, v)| \geq t \\ 0 & \text{otherwise} \end{cases}$$

Image restoration by inverse filtering

Full (i.e., no
lowpass
filter)

$$\hat{f}(x, y)$$



Lowpass
filter
limited to
radius of 40

$$\hat{f}(x, y)$$



Lowpass
filter
limited to
radius of 70

$$\hat{f}(x, y)$$



Lowpass
filter
limited to
radius of 85

$$\hat{f}(x, y)$$

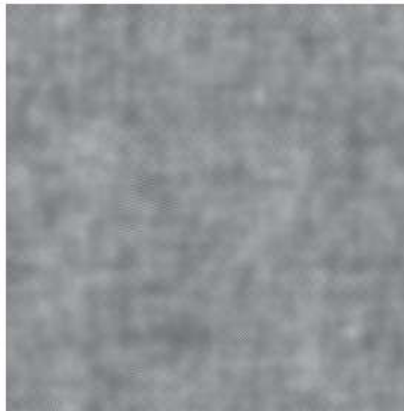


Image Deconvolution: Additive Noise

$$\mathbf{G}[k_1, k_2] = \mathbf{H}[k_1, k_2] \mathbf{F}[k_1, k_2] + \mathbf{V}[k_1, k_2] \quad \mathbf{F}[k_1, k_2] = \frac{\mathbf{G}[k_1, k_2]}{\mathbf{H}[k_1, k_2]} - \frac{\mathbf{V}[k_1, k_2]}{\mathbf{H}[k_1, k_2]}$$

In many practical applications, $\mathbf{H}[k_1, k_2]$ may assume very small values for large values of $[k_1, k_2]$. Consequently, the second term in Eq. (6.28) may end up *amplifying* the noise component and

So instead of using $\mathbf{F}[k_1, k_2] = \frac{\mathbf{G}[k_1, k_2]}{\mathbf{H}[k_1, k_2]} - \frac{\mathbf{V}[k_1, k_2]}{\mathbf{H}[k_1, k_2]}$ minimize the Tikhonov cost criterion

$$e = \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} [(g[n, m] - h[n, m] * \hat{f}[n, m])^2 + (\lambda \hat{f}[n, m])^2]$$

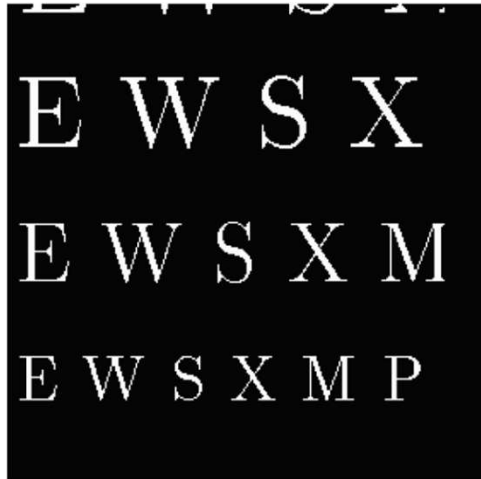
How do we find the $\hat{f}[n, m]$ that minimizes e ? Use a Wiener filter (next slide).

Image Deconvolution: Wiener Filter

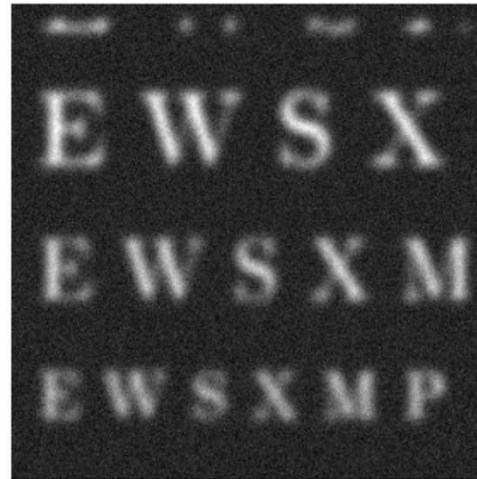
$$E = \frac{1}{N^2} \sum_{k_1=0}^{N-1} \sum_{k_2=0}^{N-1} [|\mathbf{G}[k_1, k_2] - \mathbf{H}[k_1, k_2] \hat{\mathbf{F}}[k_1, k_2]|^2 + \lambda^2 |\mathbf{X}[k_1, k_2]|^2]$$

The error can be minimized separately for each (k_1, k_2)

using a Wiener filter: $\hat{\mathbf{F}}[k_1, k_2] = \mathbf{G}[k_1, k_2] \frac{\mathbf{H}^*[k_1, k_2]}{|\mathbf{H}[k_1, k_2]|^2 + \lambda^2}$

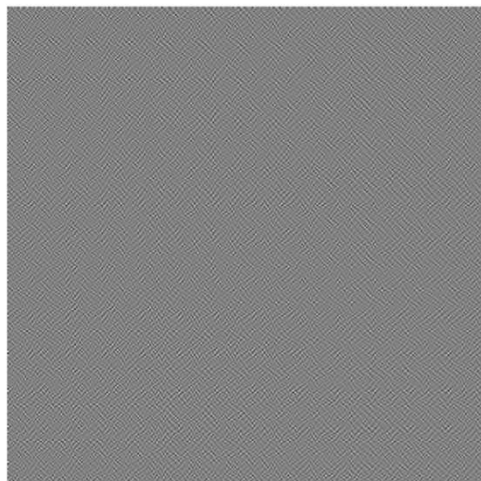


(a) $f[n,m]$



(b) $g[n,m] = h[n,m] ** f[n,m] + v[n,m]$

Without Tikhonov Regularization, get garbage, since noise is amplified and drowns out the reconstructed image.



(c) $\hat{f}_1[n,m]$ without regularization

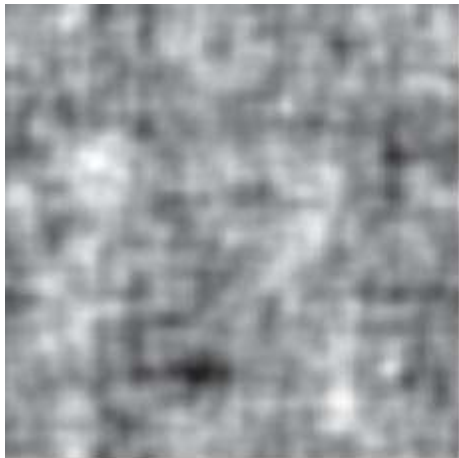


(d) $\hat{f}_2[n,m]$ with Wiener filter

With Tikhonov regularization, when implemented using a Wiener filter, produces a recognizable image.

Image restoration by Wiener filtering

Inverse filtering



Full

$$\hat{f}(x, y)$$



Lowpass filter
radially limited

$$\hat{f}(x, y)$$

Wiener filtering

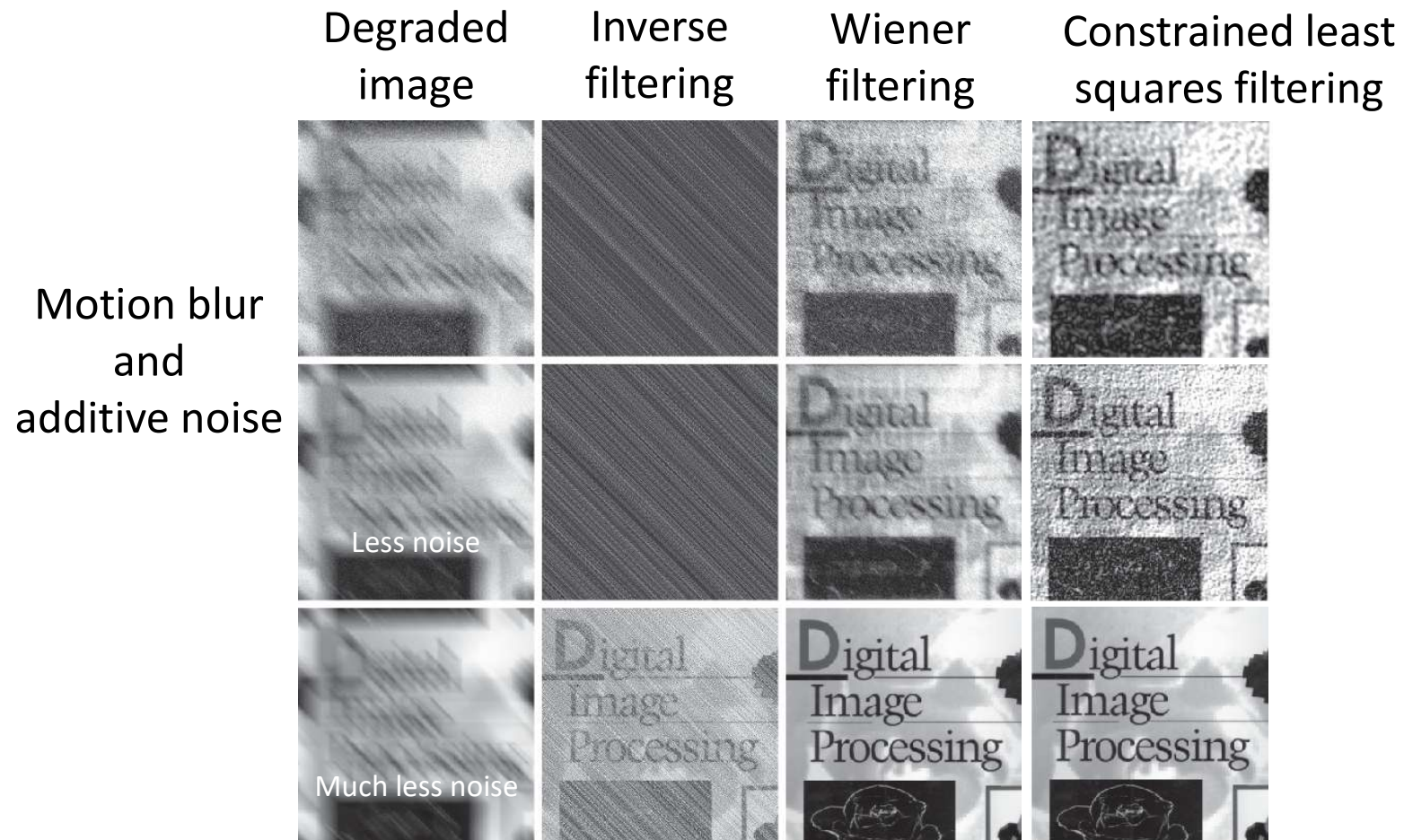


$$\hat{f}(x, y)$$

$$e^2 = E\{(f - \hat{f})^2\}$$

Minimum mean
square error filter

Image restoration by constrained least squares filtering



Motion Blur





Assumptions – Motion Blur

- The blurred image has been appropriately *rotated* so that the direction of motion is aligned along the x axis.
- *Image recording* is T (seconds) in duration, with instantaneous start and stop actions, thereby allowing us to represent the process in terms of a rectangle function.
- The motion of the imager, relative to the scene, is linear (along the x axis) and at a *constant speed* s (meters/s).

Motion-Blur Deconvolution: Formulation

Continuous-time

$$g(x, y) = \int_0^T f(x - x', y) dt$$

$$g(x, y) = f(x, y) * h(x, y). \quad h(x, y) = \frac{1}{s} \text{rect} \left(\frac{x - D/2}{D} \right) \delta(y)$$

$$\mathbf{H}(\mu, \nu) = \mathcal{F}_{x \rightarrow \mu} \left\{ \text{rect} \left(\frac{x - D/2}{D} \right) \right\} \mathcal{F}_{y \rightarrow \nu} \left\{ \frac{\delta(y)}{s} \right\} = \frac{D}{s} \text{sinc}(\mu D) e^{-j\pi\mu D}$$

$$\begin{aligned} \mathbf{H}(\Omega_1, \Omega_2) &= \text{DTFT}_{n \rightarrow \Omega_1} \left\{ \text{rect} \left[\frac{n - N/2}{N/2} \right] \right\} \text{DTFT}_{m \rightarrow \Omega_2} \{ \delta[m] \} \frac{T}{N} \\ &= \frac{T}{N} \frac{\sin \left[\Omega_1 \left(\frac{N+1}{2} \right) \right]}{\sin(\Omega_1/2)} e^{-j\Omega_1 N/2} \end{aligned}$$

Discrete-time

Courtesy of Yagle and Ullaby

$$\mathbf{G}(\Omega_1, \Omega_2) = \mathbf{F}(\Omega_1, \Omega_2) \mathbf{H}(\Omega_1, \Omega_2). \quad (6.48)$$

Here, $\mathbf{G}(\Omega_1, \Omega_2)$ is the 2-D spectrum of the recorded blurred image, and $\mathbf{H}(\Omega_1, \Omega_2)$ is the discrete-space spatial frequency response function (DSSF) response of $h[n, m]$. From entry #7 in **Table 2-8**, and noting that $\Delta_t = T/N$, the DSSF response function corresponding to Eq. (6.47) is given by

$$\begin{aligned} \mathbf{H}(\Omega_1, \Omega_2) &= \text{DSFT}\{h[n, m]\} \\ &= \text{DTFT}_{n \rightarrow \Omega_1} \left\{ \text{rect} \left[\frac{n - N/2}{N/2} \right] \right\} \\ &\quad \times \text{DTFT}_{m \rightarrow \Omega_2} \{ \delta[m] \} \frac{T}{N} \\ &= \frac{T}{N} \frac{\sin \left[\Omega_1 \left(\frac{N+1}{2} \right) \right]}{\sin(\Omega_1/2)} e^{-j\Omega_1 N/2}. \end{aligned} \quad (6.49)$$

The sinc function dictates that $\mathbf{H}(\Omega_1, \Omega_2) = 0$ for $\Omega_1(N+1)/2 = k\pi$ for nonzero integer values of k . Consequently, the inverse filter $1/\mathbf{H}(\Omega_1, \Omega_2)$ is undefined at these values of Ω_1 , thereby requiring regularization.

Wiener Filter Solution

$$\begin{aligned} \mathbf{F}(\Omega_1, \Omega_2) &= \frac{\mathbf{G}(\Omega_1, \Omega_2) \mathbf{H}^*(\Omega_1, \Omega_2)}{|\mathbf{H}(\Omega_1, \Omega_2)|^2 + \lambda^2} \\ &= \frac{\mathbf{G}(\Omega_1, \Omega_2) \frac{T}{N} \frac{\sin \left[\Omega_1 \left(\frac{N+1}{2} \right) \right]}{\sin(\Omega_1/2)} e^{j\Omega_1 N/2}}{\left(\frac{T}{N} \right)^2 \left| \frac{\sin \left[\Omega_1 \left(\frac{N+1}{2} \right) \right]}{\sin(\Omega_1/2)} \right|^2 + \lambda^2}. \end{aligned} \quad (6.50)$$

Example of **motion-blur deconvolution**.
Wiener filter used to reconstruct sign
from noisy motion-blurred image.



(b) Image $g'[n,m]$: motion-blurred highway sign
with additive noise



(c) Reconstructed highway sign